My Great Project in 8 Minutes

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Introduction

- This is the physics problem/topic I studied
- This is why this project is interesting
- There should be 8-10 slides and backup
- The font size should in general be >= 24
- See also the slides idea_ResearchPresentation-lucia-depaul.ppt in the info folder

Introduction

- Introduce the physics problem/topic you are studying
- Introduce the physics at a level the class will understand

Method

- This is an overview of the work I did on my project
- These are the important equations

$$E = mc^2$$

- These are the computational methods I used to solve/study this equation

Method

- These are some of the problems I encountered in my project and how I overcame them (optional)
- How I verified my code was correct
- The computational method used in this study does / does not overlap (either ok) with what we covered in PHYS 305 and here is a summary of it

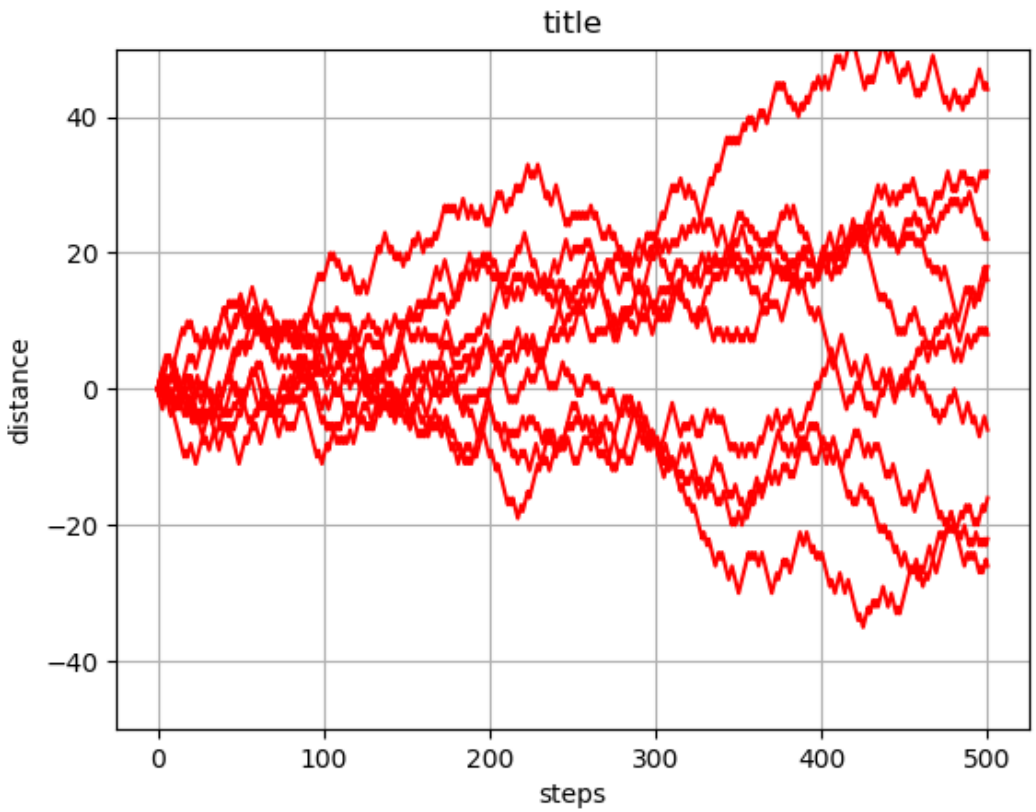
Method

- DO NOT SHOW CODE

```
# move the balls according to their velocity
            x[i] = x[i] + vx[i]*h
            y[i] = y[i] + vy[i]*h
# if at boundary, reflect from the walls
            if x[i] < 0:
               x[i] = 0.
               vx[i] = -vx[i]
            if x[i] > length:
               x[i] = length
               vx[i] = -vx[i]
               y[i] < 0:
               i] = 0.
                  = -vy[i]
                  gth:
      for i in
                                                  i])**2) < eps2:
                  = 0
                  y_cm = 0.5*(
                  u1x = vx[i] - vx
                  u1y = vy[i] - vy_cm
                  umag = math.sqrt(u1x**2 + u1y**2)
                  theta = 2*np.pi*random.random()
                  vx[i] = math.cos(theta)* umag + vx_cm
                  vx[j] = -math.cos(theta)* umag + vx_cm
                  vy[i] = math.sin(theta)* umag + vy_cm
                  vy[j] = -math.sin(theta)* umag + vy_cm
```

Results

- These are my results. And note the graph is labeled.



Results

- Here are some additional results. I got these by changing initial conditions, other parameters, ...
- Or in cases of animation, perhaps you will show the animation under different conditions, leading to different results

Analysis

- My results showed this aspect of physics. We gained this insight into this problem
- By varying the initial conditions, I also learned this.
- The errors associated with my method are the following.

Conclusions and Future Work

- Here is a line of conclusions
- Here is a second line of conclusions
- Here are some ideas for future work based on what I learned in this project

Backup

- Good idea to have one or two backup slides for questions
- Contains additional background information to the problem

Backup

- Or could contain some additional results that you did not have time to show

- Always keep to your time limit
  - You will have to practice your talk a couple times to best gauge the time and the number of slides
  - Unlike this slide, try to limit the number of bullet points, try to limit the number of words

Analysis

- I was also expecting to see X but instead I observed Y. I don't fully understand the result. It could be my code, or it might be because of this approximation. Here are some things I tried to resolve this.